Measurement 258 (2026) 119256

Contents lists available at ScienceDirect

Measurement

journal homepage: www.elsevier.com/locate/measurement

Reverse digital twin mapping of building envelope anomalies based on UAV infrared point clouds

Wenjie Shen [a], Chen Wang [a,*], Can Zhu [a], Defa Wang [b], Zhibin Chen [c]

[a] Intelligence and Automation in Construction Fujian Province Higher-educational Engineering Research Centre, College of Civil Engineering, Huaqiao University, 361021 Xiamen, China
[b] China Railway No. 18 Bureau Group No. 1 Engineering Co., Ltd., Zhuozhou District, Baoding 072750, China
[c] China Construction Third Engineering Bureau Group South China Co., Ltd., Guangzhou 510080, China

ARTICLE INFO

Keywords:
Infrared point cloud technology
Infrared point cloud fusion
Structural anomaly detection
Digital twin mapping

ABSTRACT

As the service life of some buildings increases, some safety hazards may seriously endanger public safety and cause economic losses. Defect detection and management are necessary steps to eliminate potential safety hazards. At present, the inspection of building exterior walls mostly relies on manual inspection, which is highly subjective, risky and has low inspection accuracy. To this end, this study developed an algorithm based on the infrared and point cloud data fusion model of the Point Cloud Reconstruction Network (PCRNet) to increase the depth information of the model and improve its understanding of complex scenes. For the anomaly detection of regular and irregular geometric envelope structures, an automatic anomaly detection algorithm model of geometric envelope structures based on convolutional neural network (CNN) is proposed. By comparing the extracted temperature information with the labeled data in the real abnormal data set, the detection accuracy, recall rate and accuracy of the evaluation model have been improved. Finally, a reverse digital twin mapping method based on the abnormal information of spatial geometric envelope structure of infrared point clouds is proposed, which is convenient for systematic and object-oriented management. The experimental results show that among 299 experimental data points, single-data anomaly detection can be completed quickly. The average accuracy rate of various hazard detection is 92%, the Precision is 96.3 %, the Recall Rate is 94 %, the F1-Score is 95.1 %, and the average detection speed is 5.21 s. It only takes 26 min to complete the building anomaly detection.

1. Introduction

With the rapid development of the social economy, the construction industry is also booming, and people's demand for the functionality and aesthetics of buildings is growing day by day. However, with the increase in the service life of the building, hidden security risks are gradually exposed, such as cracks and external walls falling off, and other problems may seriously threaten public safety and cause huge economic losses[1,2]. Therefore, it is crucial to conduct safety inspections and manage external wall defects promptly to prevent potential safety hazards, casualties, and economic losses. Currently, the majority of building envelope inspections are manual, requiring inspectors to work at heights and manually identify defects [3]. This manual inspection method has high subjectivity, high risk, and low precision, resulting in low efficiency in building inspection and maintenance. Using visible spectrum images to create 3D models may not detect all damage. In addition, certain building areas are difficult to reach, such as the roofs and facades of irregular high-rise buildings. To make manual inspection safer and cheaper, to look into and learn more about more things, and to solve the problem of finding and managing defects in buildings with odd conditions, we need to create a highly automated method based on spatial geometry envelope. The possibility of combining heat loss detection with unmanned aerial vehicles (UAVs) simplifies the inspection of hard-to-access or dangerous areas [4]. It is a common detection method to use a thermal imager to capture data remotely.

At present, infrared thermography (IRT) has been employed in diagnostic applications across various scientific domains, such as

* Corresponding author.
E-mail addresses: 23013086022@stu.hqu.edu.cn (W. Shen), wch@hqu.edu.cn (C. Wang), 21013086040@stu.hqu.edu.cn (C. Zhu), 85852526@qq.com (D. Wang), 1109345781@qq.com (Z. Chen).

https://doi.org/10.1016/j.measurement.2025.119256
Received 18 March 2025; Received in revised form 4 September 2025; Accepted 6 October 2025
Available online 11 October 2025
0263-2241/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

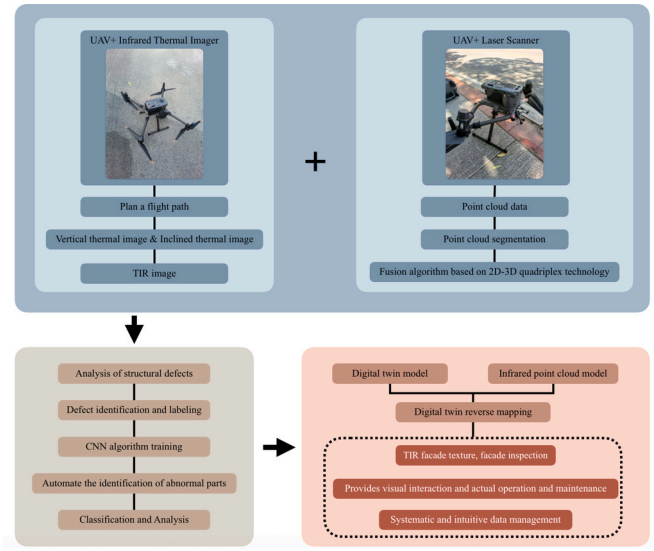


Fig. 1. General technical flow chart.

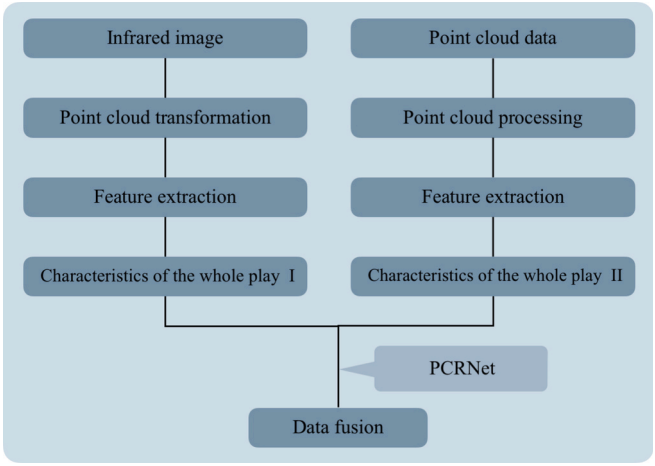


Fig. 2. Basic flow of fusion of image data and point cloud data.

medicine, agriculture, and building science[5–9]. Infrared (IR) imaging is dependent on the detection of blackbody radiation. In building environment surveys, infrared imaging has been utilized for various functions, for instance, thermal bridge detection, moisture characterization, and envelope defect identification[10,11]. Although infrared detection technology can reduce the detection time, it has certain limitations as infrared imaging is unable to provide structural information inside or deep within the building. Compared with visible light images, the resolution of infrared images is lower[12,13]. Consequently, a single method may perform inadequately, resulting in inaccurate recognition and making the localization of defects a challenge[14]. For this reason, global shape information needs to be added to detect thermal defects with complex geometries[15].

With the rise of 3D reconstruction, many researchers have increased recognition accuracy by using 3D models to add depth information to image detection[16,17]. For instance, researchers reconstruct the 3D model through image superposition, overlay it with the infrared image, and add depth information by generating three-dimensional point clouds. The 3D thermal model reconstructed by these methods enables users to visualize 3D spatial information, query and analyze temperature, and thus speed up the process of building diagnosis and modeling [18,19]. Currently, most studies rely on images to generate point cloud models or 3D models. Although these methods simplify the difficulty of reconstruction, image-based modeling methods are subject to significant limitations, such as occlusion, overlap, shadow, etc.

Therefore, this study proposed a fusion model based on PCRNet, which fused building heat map collected by infrared thermal imager collected by UAV with point cloud data. And a CNN anomaly detection model is developed to detect abnormal air leakage (heat loss) in the building envelope environment, the dangerous area of the external plaster layer and the dangerous falling off of the tile, and the building envelope with irregular structure. Finally, a set of inverse digital twin mapping and management framework for building envelope anomaly information based on UAV infrared point cloud is proposed.

The following are the main contributions and innovations of this study:

(1) This study fuses infrared and point cloud data and adds corresponding depth information to solve the problem of infrared imaging detection for irregular enclosure geometries such as arcs and spheres. Through the analogical analysis of regular and irregular geometric structures, a reference data for abnormal

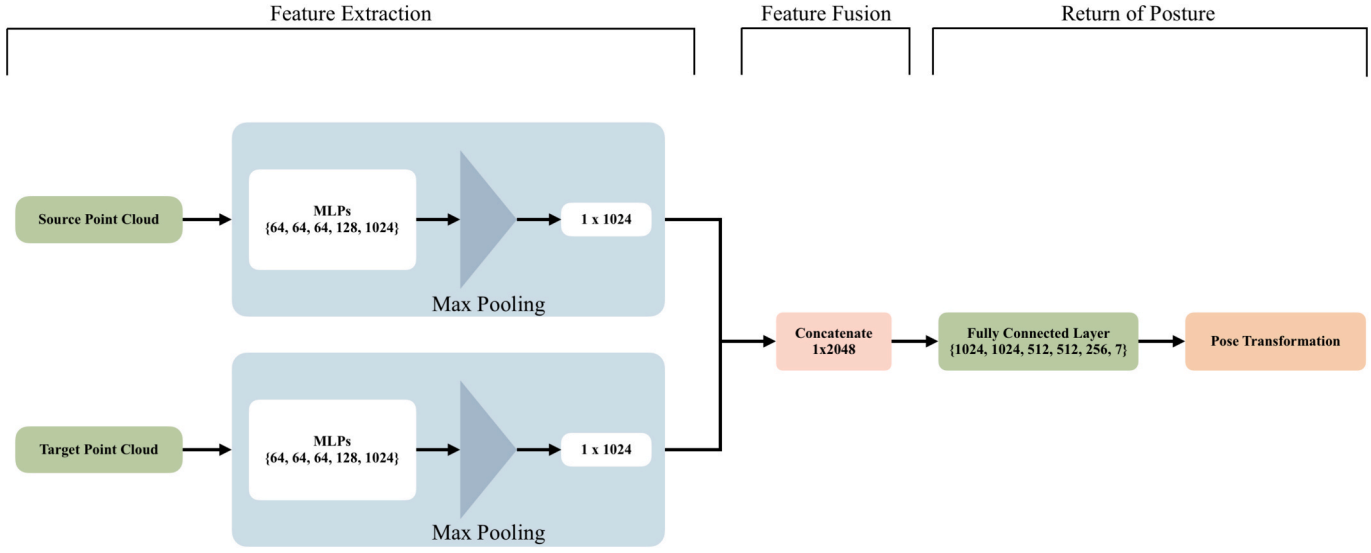


Fig. 3. Structure of The PCRNet Network Model.

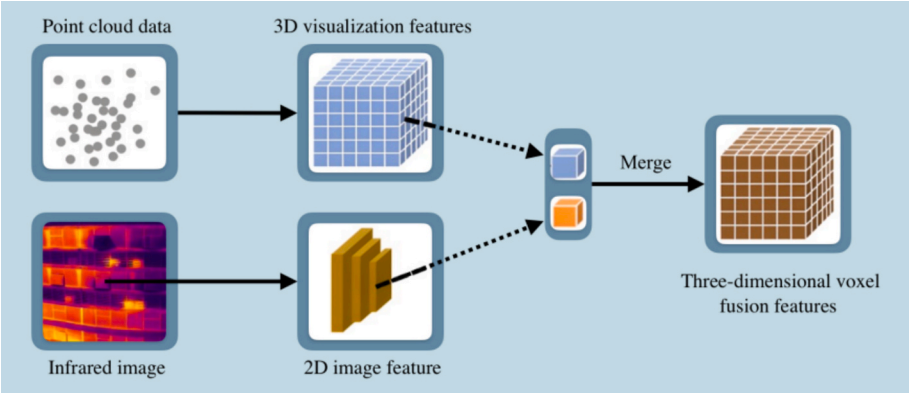


Fig. 4. Fusion Based on PCRNet.

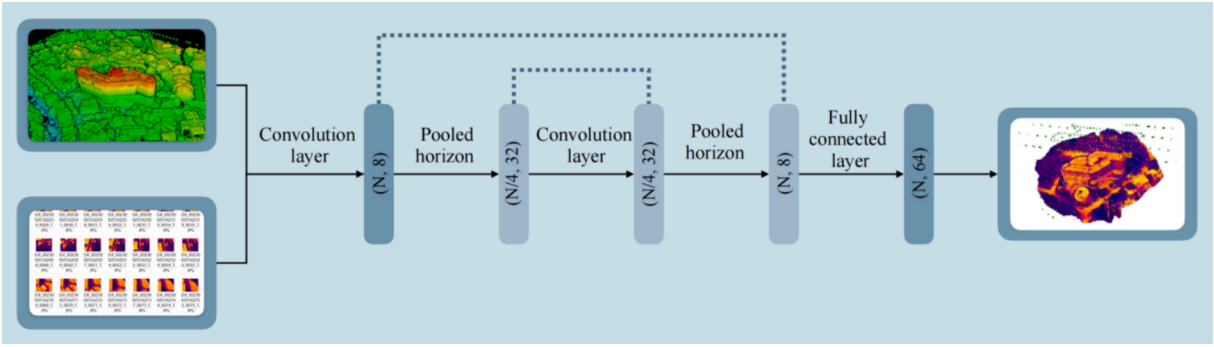


Fig. 5. Iterative PCRNet network architecture.

Table 1
PCRNet data fusion model brief code.

PCRNet data fusion model brief code

def build_fusion_model(input_shape_infrared, input_shape_point_cloud):
  # Infrared data entry
  input_infrared = Input(shape = input_shape_infrared, name='input_infrared')
  # Point cloud data input
  input_point_cloud = Input(shape = input_shape_point_cloud,
  name='input_point_cloud')
  # Define the infrared data processing layer
  conv1_ir = Conv2D(64, (3, 3), activation='relu', padding='same')(input_infrared)
  conv2_ir = Conv2D(64, (3, 3), activation='relu', padding='same')(conv1_ir)
  # Define the point cloud data processing layer
  conv1_pc = Conv2D(64, (3, 3), activation='relu', padding='same')
  (input_point_cloud)
  conv2_pc = Conv2D(64, (3, 3), activation='relu', padding='same')(conv1_pc)
  # Combining infrared and point cloud features
  merged = Concatenate(axis = -1)([conv2_ir, conv2_pc])
  # Fully connected layer
  flattened = Flatten()(merged)
  dense1 = Dense(128, activation='relu')(flattened)
  output = Dense(1, activation='sigmoid')(dense1)
  # Build model
  model = Model(inputs = [input_infrared, input_point_cloud], outputs = output)
  return model

defect detection is obtained to help inspect the building condition and promote maintenance plans or the next theoretical research.

(2) By using drones equipped with infrared imagers to inspect the exterior walls, roofs and other envelope structures of buildings, and through automated detection and management methods, the building envelope structures are inspected. This reduces the consumption of human resources and avoids manual high-altitude operations to prevent personnel hazards, achieving safe and efficient abnormal detection and management maintenance of building envelope structures.

(3) The 3D thermal model reconstructed through the fusion of point clouds and infrared thermal maps enables users to visualize, query and analyze temperature by referring to 3D spatial information, thereby accelerating the building diagnosis process and facilitating later management. This study proposes a general framework for defect detection and management of building spatial geometric envelope structures based on the combination of infrared thermal imaging and point clouds, which is also applicable to the assessment of other structures.

2. Background and related studies

2.1. Infrared thermal imaging detection

IRT is a non-contact, nondestructive evaluation NDE technology that allows the measurement of the energy emitted by the object under test (OUT) in the infrared spectrum. Balaras et al. [20] demonstrated that the surface emissivity of the OUT can produce an infrared image known as a heat map, highlighting anomalies in the temperature distribution that identify faulty parts and damaged structural parts. Kylili et al.[8]introduce pre-nondestructive testing (NDT) for building diagnosis using passive, active pulse, and active locking thermal imagers, emphasizing that while IRT is a useful tool for characterizing defects in the construction field, there is great promise for developing more advanced, efficient, and accurate methods that would employ a combination of thermal imaging methods. Feng et al.[21] propose a portable automatic detection system for quantitative corrosion detection of the replaced steel wire. Based on the discovery of the surface temperature characteristics of the corroded steel wire after heating, the automatic detection device can be upgraded to achieve quantitative detection of in-situ corrosion of the bridge cable.

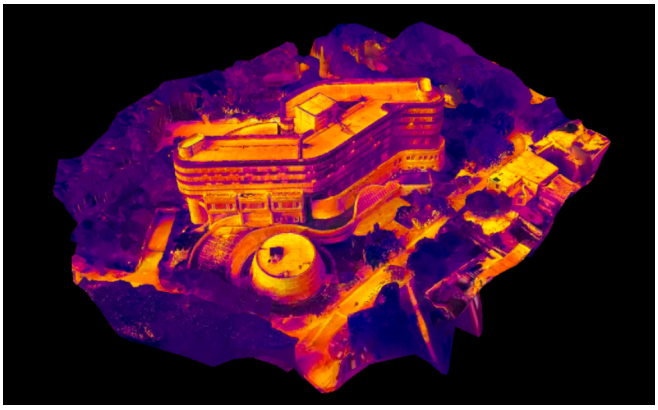


Fig. 6. Infrared Point Cloud 3D Fusion Model.

Several fields, including environmental monitoring and non-destructive evaluation (NDE) of engineering structures, have employed IRT, demonstrating its ability to detect defects with a precision that is comparable to other commonly used techniques[22]. Liu et al.[23] infrared thermal imaging and deep learning, especially CNN, were applied to classify the severity of fatigue cracks in asphalt pavement. The data set of fatigue cracking of asphalt pavement is constructed. The effect of image type on the classification of fatigue crack severity is studied, and the applicability of infrared thermal imaging to crack detection (crack severity classification and crack segmentation) is discussed. IRT can be classified as active or passive[24]. In the former, an external heat source preheats OUT and enhances detection, while the latter relies on the object's own temperature. While active IRT can better detect defects by enhancing the temperature difference between OUT and the environment, preheating structures are impractical in large-scale applications, according to Avdelidis et al. [25].

According to the above research, the use of infrared thermal imaging for building and infrastructure inspection has improved the efficiency and safety of purely manual inspection. However, due to external conditions, particularly temperature, and the lower resolution of infrared

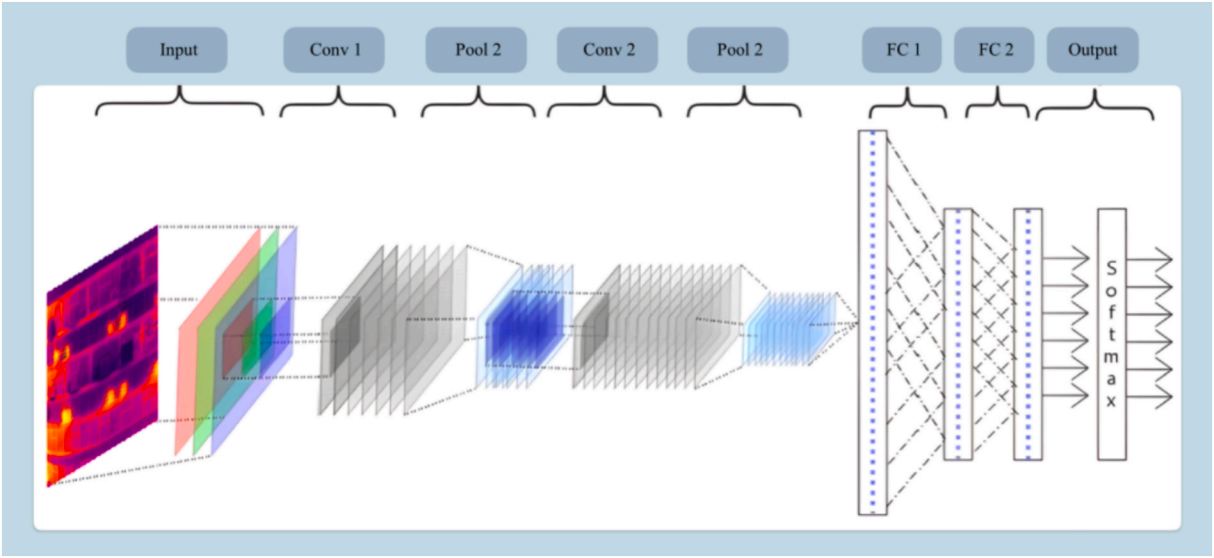


Fig. 7. Network Structure Diagram of CNN Anomaly Detection Model.

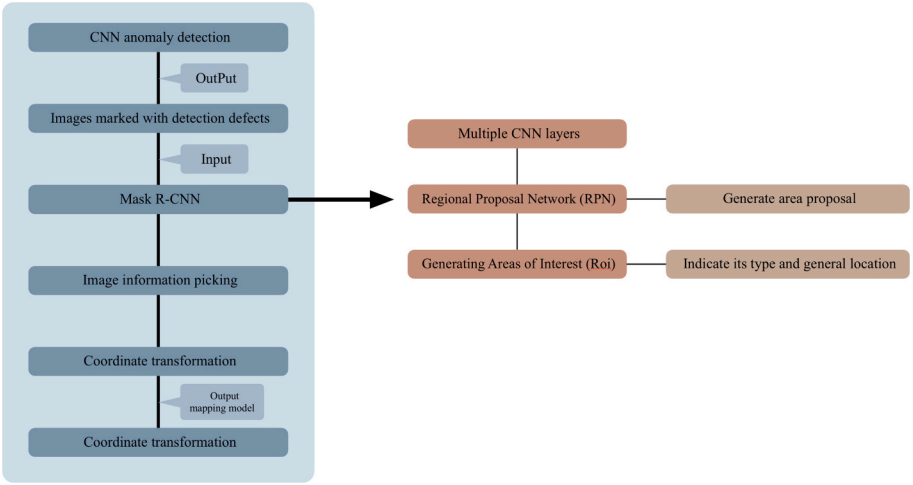


Fig. 8. Workflow of Mask R-CNN.

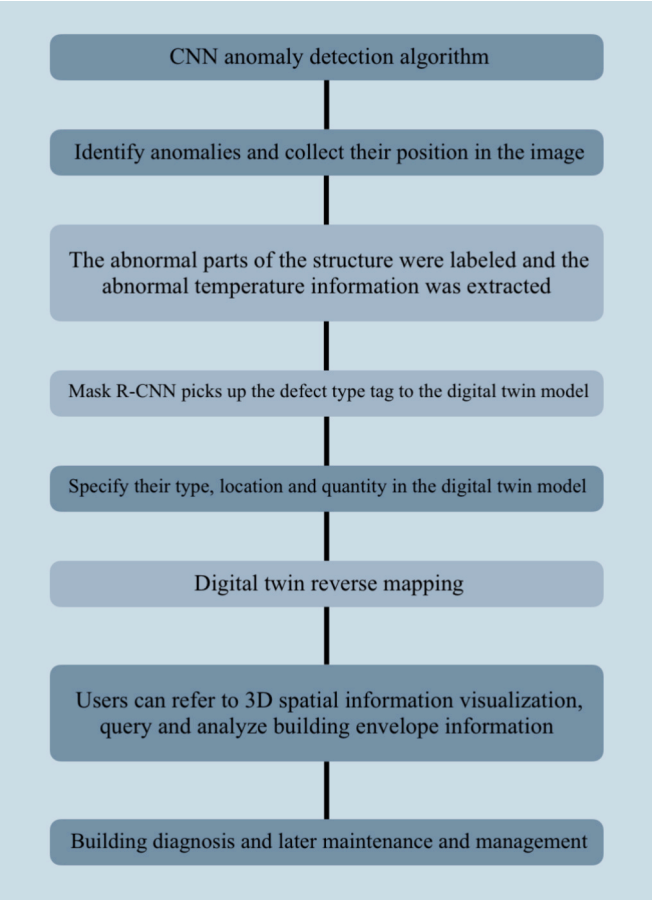


Fig. 9. General Framework for Reverse Digital Twin Mapping and Management of Anomaly Information of Spatial Geometric Envelope Based on Infrared Point Cloud.

images compared to visible images, the use of a single imaging method can perform poorly, leading to inaccurate identification and making it challenging to pinpoint the defect.

2.2. Three-dimensional point cloud technology

In the technical processing of 3D point cloud, current mainstream research combines the point cloud with other technologies, such as BIM, infrared imaging, and 3DMAX, to complete the 3D reconstruction of the target building. At the same time, a large part of the point cloud processing methods, such as point cloud noise reduction filtering[26], point cloud feature extraction[27], point cloud registration [28], and other algorithms, are also studied. In order to improve the processing efficiency of point cloud data. Tim Kaiser et al.[29] extracted 3D line segments from image data and associated them with boundary surfaces of architectural models to propose a highly automated point-cloud registration algorithm for photographic measurement that does not require the use of control points. Truong-Hong et al.[30] studied the extraction of structural components of concrete buildings from laser scanning point clouds at construction sites and proposed a method for automatically extracting surface point clouds of structural components of concrete buildings, which could be used for construction quality detection and defect inspection based on surface geometry information. Min-Koo Kim et al.[31] developed a technique based on laser scanning to automatically evaluate reinforced concrete units, including steel bar spacing and concrete cover relative to formwork, by studying the automatic assessment of reinforced concrete member formwork and steel bar size quality. Dong et al.[32] a three-dimensional point cloud enhancement and segmentation system for potholes is proposed. The system is made up of a 3DGAN transformer fused with a 3D convolution network called Conv-Trans-3DGAN and a 3D segmentation network fused with a 3DGAN transformer named Conv-Trans-3DSeg. Through ablation comparison experiments, accuracy and F1 scores went up by 4.733 % and 3.03 %, respectively.

2.3. Infrared point cloud defect detection technology

Infrared point cloud defect detection technology is a new detection method combining infrared imaging technology and three-dimensional point cloud data analysis, which is widely used in industrial inspection, building monitoring, object surface defect detection and other fields[33–35]. The core principle is to obtain the heat distribution map of the object surface by infrared sensor, and then combine the three-dimensional point cloud data for defect analysis and positioning.

In the study of infrared detection, Angeliki Kylili et al.[8]detailed the working principle of the infrared thermal imager, but the accuracy of the measurement depends on many parameters, including emissivity,

Fig. 10. Schematic Diagram of Building Envelope in This Case.

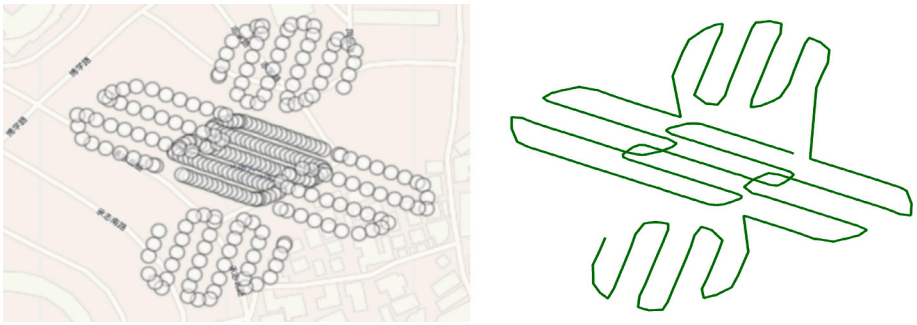


Fig. 11. UAV Flight Paths.

(a) (b)

Fig. 12. UAV Data Acquisition Equipment. (a) is DJI UAV M600; (b) is ZENMUSE XT2 Thermal Imaging Camera.

Table 2
DJI UAV M600 performance parameter information.

Serial number	Performance name	Argument
1	Maximum take-off weight	15.1 kg
2	Hover accuracy	Horizontal: ±1.5m, Perpendicularity:±0.5m
3	Hover time without load	35 min
4	Load 6 kg hover time	16 min
5	Maximum pitch Angle	25°
6	Maximum ascent speed	5 m/s
7	Maximum descent speed	3 m/s
8	Maximum wind speed	8 m/s
9	Maximum take-off altitude	2500 *m*
10	Maximum horizontal flight speed	18 m/s (Windless environment)
11	Remote control maximum communication distance (no interference, no block)	5 km(FCC mode)
12	Operating ambient temperature	−10 °C ~ 40 °C

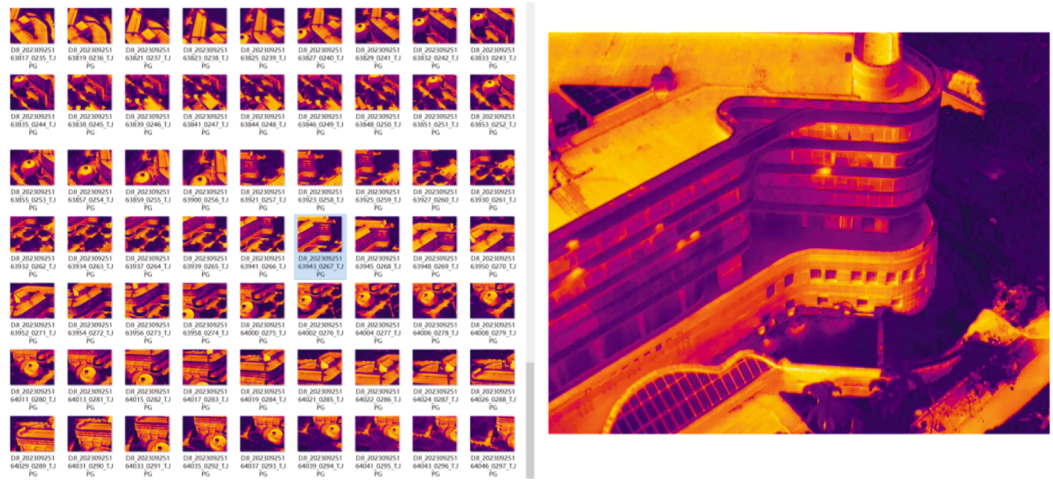
Table 3
Performance Parameters of The ZENMUSE XT2 Thermal Imaging Camera.

Serial number	Performance name	Argument
1	Pixel spacing	17 μm
2	Wavelength range	7.5 ~ 13.5 μm
3	Full frame frequency	30 Hz
4	Sensitivity (NEdt)	< 50mk @f/1.0
5	Temperature range in high-gain scenarios	640–512:-25℃ ~ 135℃ ; 336-256:-25℃ ~ 100℃
6	Temperature range for low-gain scenarios	−40℃ ~ 550℃
7	Pitch tilt range	+45° ~ −130°
8	Horizontal rotation range	±330°
9	Rolling range	−90° ~ +60°
10	Support picture format	JPEG、TIFF, R-JPEG and Visible Camera: JPEG

atmospheric particles, ambient temperature, wind speed, and distance from the object under investigation, which can lead to incorrect readings. Ma et al.[36] propose that point cloud reconstruction technology is increasingly important in many engineering fields because it can generate 3D representations of objects, grids, and geometric models, and quantify surface degradation. Siebert et al.[4] analyzed the possibility of combining loss detection with unmanned aerial vehicles (UAVs) to simplify the inspection of hard-to-access or hazardous areas. Ali et al.[37]used a low-power thermal sensor and a drone to detect heat loss on building roofs and high-rise facades. Marco Puliti et al.[33] analyzed the digital 3D model obtained directly from infrared images used as inputs for the structure from motion (SfM) algorithm. The conversion of infrared spectral images to RGB images preserves the robustness of the SIFT operator[2], eliminating the need to extract contours and register IR and visible images to generate high-resolution virtual models of OUT [38,39].

In 2022, Dabetwar et al. combined infrared thermal imaging (IRT) with UAVs and then utilized point cloud reconstruction, which can be used to generate structural renderings that highlight areas of excessive heat loss. For the first time, a procedure for developing structures using SfM methods has been standardized by generating a 3D PC model using only infrared images, and the proposed model can be used to determine where heat is lost from a structure[40]. However, the proposed method can only simplify the PC reconstruction process by using pure IR images, so it has important research significance for quantifying the energy loss of structures by using 3D models. Huang et al.[18] combined point cloud and thermal imaging to generate 3D models to identify defects in building exterior walls and used image segmentation to cluster pixels with similar surface temperatures so that the processed thermal imprisonment could be composed of limited groups. Tommaso et al.[41] propose a novel automatic multi-stage model that uses YOLOv3 network and computer vision technology to detect panel defects on aerial images

(a)



(b)

Fig. 13. Image Data. (a) Image Data for Generating Point Clouds; (b) Infrared Image Data.

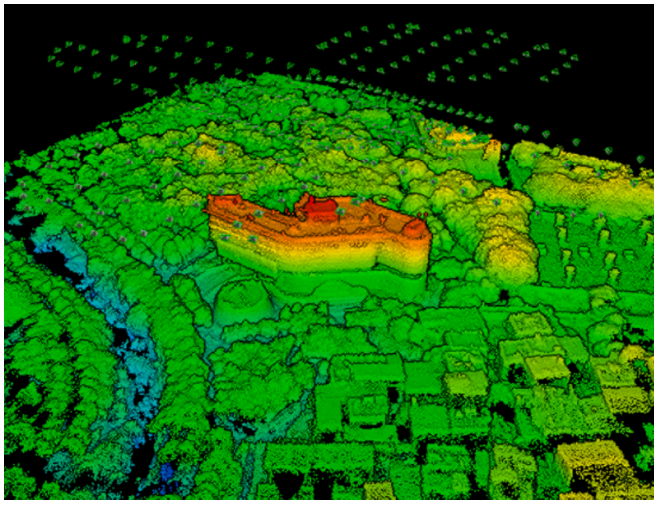


Fig. 14. 3D Point Cloud Model Generated by UAV Aerial Photography.

Fig. 15. CloudCompare Thin Point Cloud.

taken by drones. The model combines panel and defect detection to improve its accuracy, and it has the flexibility to process thermal or visible images and detect a wide variety of defects, providing operations personnel with recommended actions to provide more effective data-

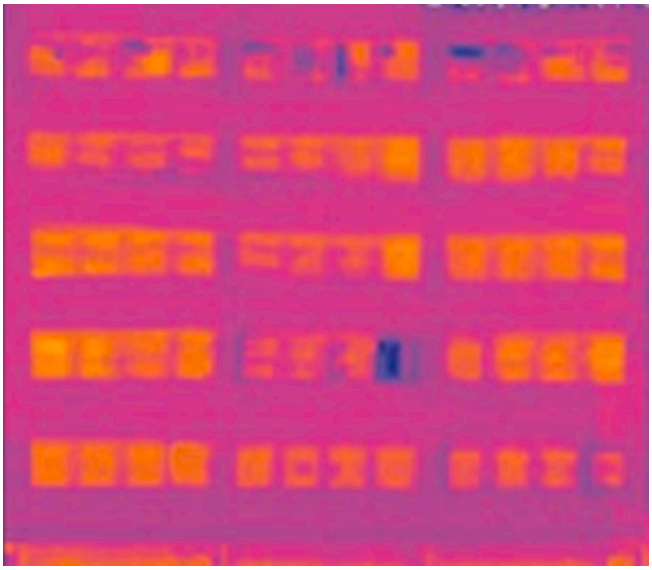


Fig. 16. Display Diagram after Gaussian Filter Processing.

driven maintenance strategies.

Many researches focus on the fusion of image modeling and infrared thermal imaging to reconstruct 3D point cloud models from the surface details provided by images and thermal imaging data. This method can effectively combine thermal information with geometric structures, but it usually relies on 2D images as an intermediary, and there is a lack of research directly using 3D point cloud data. Therefore, by introducing UAV 3D point cloud data to fusion with infrared imaging images, direct use of 3D point cloud fusion can more accurately restore the object's geometric structure and defect distribution. Combined with thermal information provided by infrared imaging, defects (such as surface cracks or deep defects) can be identified more comprehensively. Thus, the problem of infrared imaging detection for complex and irregular maintenance structures is solved.

2.4. Research based on deep learning and BIM-based management

In recent years, deep learning has achieved outstanding results in many fields of research, such as healthcare, machinery, and finance. [42–44] Moreover, deep learning has the advantages of strong learning ability, wide coverage, strong adaptability, and good portability in construction engineering image detection, especially in the automatic inspection of buildings, combining 3D point cloud and infrared thermal imaging technology, and training data based on deep learning methods to establish an automated inspection system. It has great application prospects in the management of building operations and maintenance stages. Omar et al.[12] proposed that the application of the IRT and K-means clustering algorithms showed good results in the automatic detection and classification of layered defects of concrete bridge floors. Sang et al.[45] Deep learning and convolutional neural network methods for damage identification. Some advantages of these methods include noise elasticity while extracting features that represent damage during training without the need for pre-processing and post-processing steps[46]. For example, some scholars have proposed that deep learning and neural networks in IRT have been used to classify damage by analyzing the cooling curves of two stainless steel plates with air, oil, and water penetration[47,48]. For engineering applications, Ali et al. [49] proposed a deep learning-based method for detecting underground damage of steel truss bridges from infrared images and showed an overall accuracy of about 98 % in detecting defects compared to ultrasonic pulse velocity testing.

In view of certain risks in the detection of building envelope structures, infrared imaging technology and three-dimensional point cloud technology have advantages and disadvantages in different application scenarios. Based on the low resolution of infrared imaging images and the advantages of three-dimensional point cloud that can increase the depth information, the combination of infrared and point cloud is chosen to conduct deep learning research on the abnormal detection of building structures. Since most of the detected defect information is only

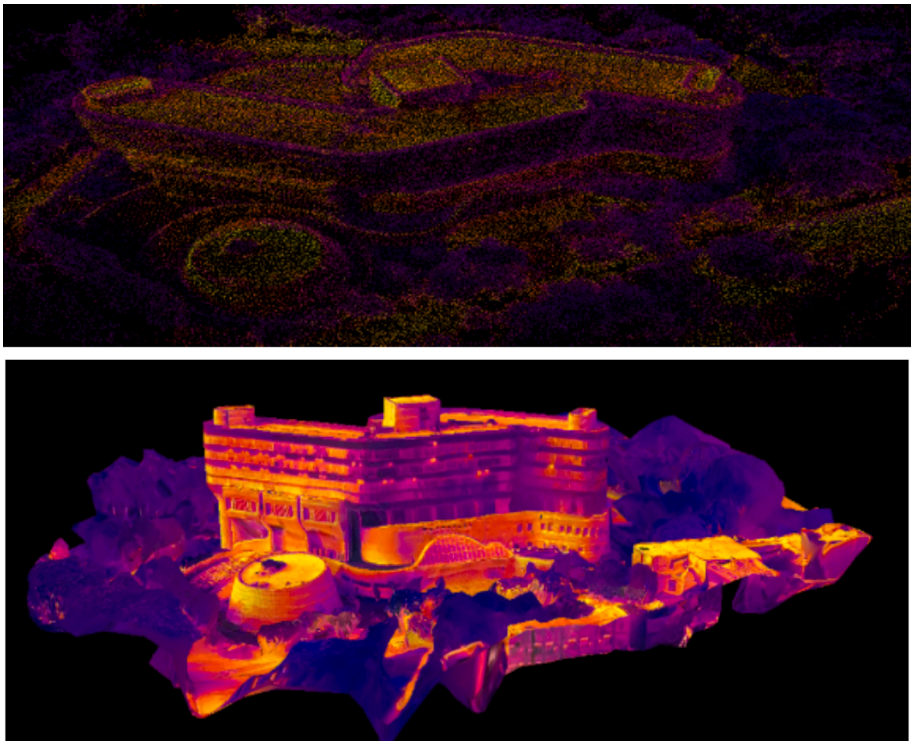


Fig. 17. Infrared Point Cloud 3D Fusion Model.

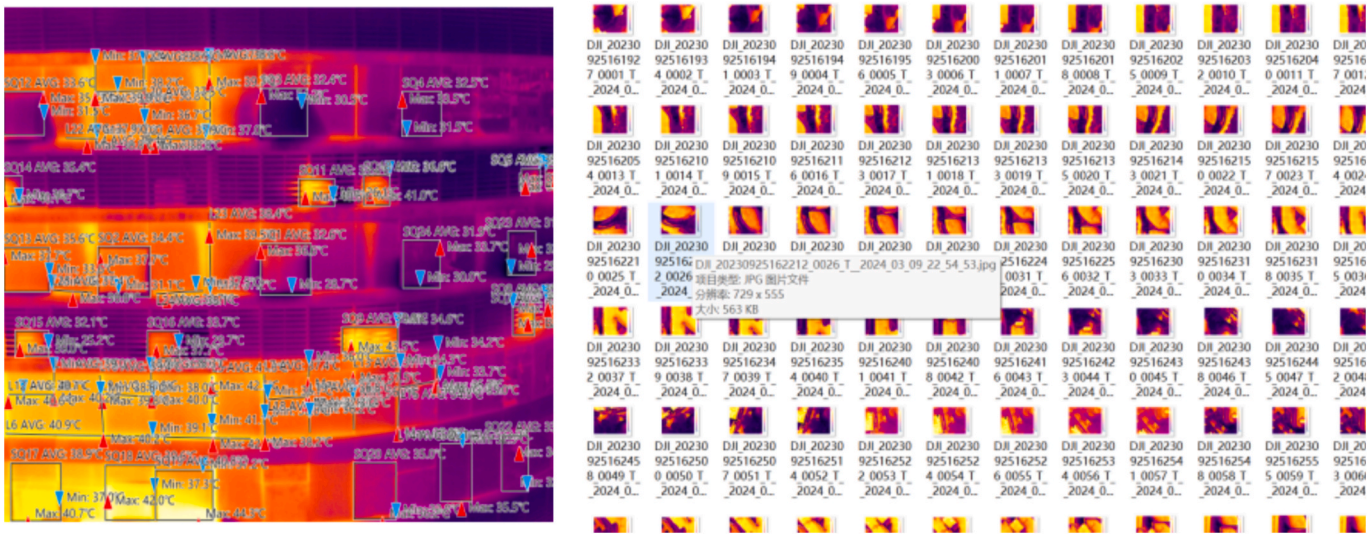


Fig. 18. DJI Thermal Analysis Tool 3 Marks Exceptions.

used to generate reports rather than stored in the digital model for systematic and object-oriented management, this study reversely maps the detected information to the digital twin model, which is more conducive to the maintenance of equipment and the automatic management of structural anomalies. In view of the current technological deficiencies, the key scientific issues addressed in this study are as follows:

(1) Address the issue of inaccurate detection when performing anomaly detection on irregular geometric maintenance structures (such as arcs, spheres, and other spatial planes) due to the lack of depth information in infrared imaging technology, which is prone to generating outliers.

(2) Addressing the issues of insufficient detection accuracy, recall rate and precision of the abnormal information detection and evaluation model, and enhancing the degree of automated image detection, it reduces the consumption of human resources and avoids manual high-altitude operations to prevent personnel hazards.

(3) Address the issue that most of the detected defect information is only used to generate reports rather than being stored in digital models for systematic and object-oriented management.

3. Methodology

In this study, the infrared thermal image and point cloud data of the building were obtained by UAV in the data collection stage, and preliminary positioning was carried out. Then, in the construction stage of detection framework, infrared image resolution is improved by fusion with RGB image, anomaly detection based on CNN is developed, and point cloud data is fused to obtain more accurate detection results. Next, the digital twin reverse mapping technology is used to improve the abnormal diagnosis, maintenance and management of building structures. The overall technical flow chart of this study is shown in Fig. 1.

This section will explain the proposed data fusion model based on PCRNet, anomaly detection model based on CNN and digital twin reverse mapping technology respectively.

3.1. Construct data fusion model based on PCRNet

PCRNet network is a kind of fully connected neural network structure with strong nonlinear mapping ability. As a deep learning model integrating point cloud data and RGB image data, it shows superior performance in object recognition and scene understanding, and reveals the possibility of building a fusion model of infrared data and point cloud data. In 3D reconstruction tasks, network structures such as PointNet are used to process point cloud data, enabling the alignment and blending of prediction results with the original image. In addition, more complex network structures and loss functions can be used to further enhance the fusion effect. PCRNet is divided into two parts, the first part is basically equivalent to the PointNet feature extraction structure, which is composed of several Multilayer Perceptron (MLP), and the data dimension is transformed into (64,64,64,128,1024). In the end, each point is transformed into a feature vector of 1024 dimensions, and then the maximum pool is done, and the maximum value is taken for each dimension, so that the point cloud of n*3 becomes a feature vector of 1*1024. The registration input is two point clouds, so you get two vectors of 1024 dimensions, and when you combine the two, you get a vector of 2048 dimensions. At this point, the PointNet-based coding process is complete.

In this study, image data is used to assist the processing of point cloud data, and local feature integration module (LFA) is added to the subsequent point cloud feature extraction for local feature extraction. The following sections describe the PCRNet model in detail.

3.1.1. Pcrnet data fusion model

In this study, an infrared camera was used as a source of raw infrared images and point cloud data. Prior to this, this study first converted infrared images into point cloud data, which was then input into the PCRNet network along with the point cloud data. Fig. 2 shows the basic flow of infrared point cloud data fusion based on PCRNet proposed by this research.

The overall framework of the PCRNet network model is shown in Fig. 3. The overall structure of this network model consists of a feature extraction module, a feature fusion module and a pose regression module. In the feature extraction module, this model uses PointNet to extract global information from the input target point cloud and source point cloud respectively. That is, it uses 5 layers of MLP to perform feature abstraction and extraction on the input point cloud, and then performs dimensionality reduction operations on the point cloud features abstracted to the high-dimensional space through 1 layer of Max pooling, thereby ensuring that the order of the input point cloud remains unchanged. In the feature fusion module, two 1×1024-dimensional global features are concatenated into one 1×2048-dimensional feature. In the pose regression module, this model uses fully connected layers of different dimensions to predict relative poses and obtain rotation and translation amounts. The fully connected layer has a total of 5 hidden layers, with dimensions of 1024, 1024, 512, 512, and 256 respectively.

Table 4
Extraction of Abnormal Labeled Temperature Information.

Monitoring point	Minimum temperature (℃)	Average temperature (℃)	Maximum temperature (℃)
SQ1	28.7	32.6	36.6
SQ2	31.1	34.4	37.7
SQ3	30.5	32.4	34.2
SQ4	31.5	32.5	33.5
SQ5	29.8	32.9	36.0
SQ6	33.2	35.4	37.5
SQ7	32.5	35.6	38.7
SQ8	33.0	35.4	37.7
SQ9	34.6	39.0	43.5
SQ10	31.8	36.4	41.0
SQ11	26.1	33.4	40.6
SQ12	31.5	33.6	35.7
SQ13	33.6	35.6	37.7
SQ14	30.7	35.4	40.1
SQ15	25.2	32.1	39.0
SQ16	29.7	33.7	37.7
SQ17	37.0	38.9	40.7
SQ18	37.3	39.6	42.0
SQ19	37.2	40.8	44.3
SQ20	33.8	35.0	36.2
SQ21	32.3	33.9	35.5
SQ22	32.1	33.2	34.3
SQ23	29.3	31.4	33.6
SQ24	30.0	31.9	33.7
L1	37.8	38.8	39.7
L2	38.2	38.5	38.8
L3	38.2	38.7	39.2
L4	37.7	38.4	39.0
L5	41.1	41.8	42.5
L6	39.1	40.9	42.7
L7	36.6	37.4	38.2
L8	36.5	38.3	40.0
L9	38.0	38.9	39.8
*L*10	38.6	39.4	40.2
*L*11	37.7	39.0	40.2
L12	39.7	40.1	40.6
L13	36.7	37.1	37.5
L14	34.3	35.0	35.8
L15	33.7	34.8	35.8
L16	34.2	35.0	35.7
L17	36.0	37.4	38.8
L18	36.7	37.3	37.8
L19	36.2	36.8	37.3
L20	36.7	37.5	38.2
L21	37.0	37.9	38.8
L22	37.0	37.9	38.8
L23	37.2	38.4	39.5
L24	37.5	38.1	38.7
L25	36.7	37.4	38.0

Table 5
Training Parameter Information of CNN Model.

Optimizer	Initial learning rate	Loss function	Batch size
Adam	0.001	Softmax	16

The dimension of the output layer is 7, which is a combination of quaternions and translation vectors.

In this study, the model adopts two different evaluation indexes: one is average accuracy (Acc); The second is root mean square error (RMSE). Acc verifies the proportion of samples correctly predicted by the model in the total sample number, and RMSE is used to measure the difference between the predicted value of the model and the true value, reflecting the standard deviation of the error between the predicted value and the actual value of the PCRNet model. Eq. (1) and (2) can represent Acc and RMSE:

$$Acc = \frac{TP + TN}{TP + TN + FP + FN}\# \tag{1}$$

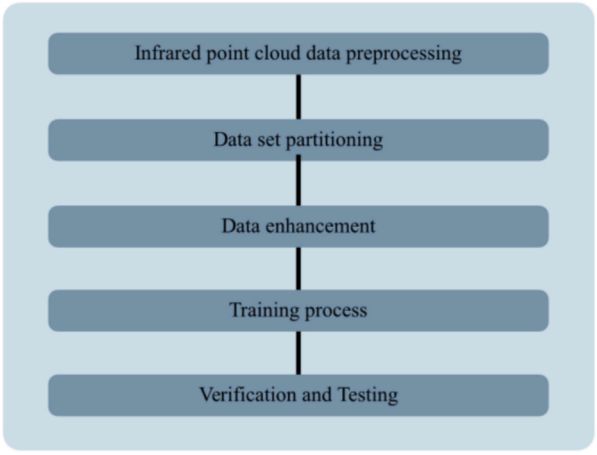


Fig. 19. CNN Abnormal Defect Detection Model Training Flow Chart.

In the Eq. (1), TP, TN, FP, and FN represent the number of true cases, true countercases, false positive cases, and false countercases respectively.

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}\# \tag{2}$$

In Eq. (2), N is the total number of samples. y_i is the true value for sample i, and $\hat{y}_i$ is the predicted value for sample i.

In this study, Principal Component Analysis (PCA) was introduced into the PCRNet model, and Adam(linear) activation function was introduced. By introducing PCA module and Adam function, Effectively improve the efficiency, accuracy and robustness of PCRNet's point cloud data processing. The PCA module ensures the high efficiency of data processing by dimensionality reduction, noise removal and extraction of important features. The Adam function, by means of adaptive learning rate and robust optimization, ensures that the model can be trained and convergent on high-dimensional, complex and noisy data. Eq. (3) represents the Adam function. The model's training times are denoted by k, the weight coefficient by g, the maximum pooling size by d, and the sampling interval by t.

$$A = k \times g \times d \times t\# \tag{3}$$

PCA module is used to help reduce the high-dimensional feature space of infrared point cloud data to a lower dimension, and Adam function is combined with adaptive adjustment of learning rate to improve training speed and optimize model performance. After the data after PCA reduction is input into PCRNet network, Acc is first used to evaluate the accuracy of infrared point cloud data fusion. The RMSE then measures the model regression performance.

In the process of training the fusion model, the binary cross-entropy loss function is selected as the model loss function in this study, As expressed by Eq. (4):

$$Loss = -\frac{1}{N}\sum_{i=1}^{N}(y_i \bullet log(\hat{y}_i) + (1 - y_i) \bullet log(1 - \hat{y}_i))\# \tag{4}$$

Where, y_i is the true label. $\hat{y}_i$ is the probability that the model predicts that the sample belongs to class 1. By minimizing the loss function, the PCRNet model learns to adjust the parameters so that the predicted probability is as close as possible to the true label.

When PCRNet builds the data fusion model, it fuses the infrared thermal image of the complex building envelope acquired by the UAV with the three-dimensional point cloud data to provide depth information for the digital twin model. Fig. 4 shows the process of PCRNet network fusing infrared point cloud data.

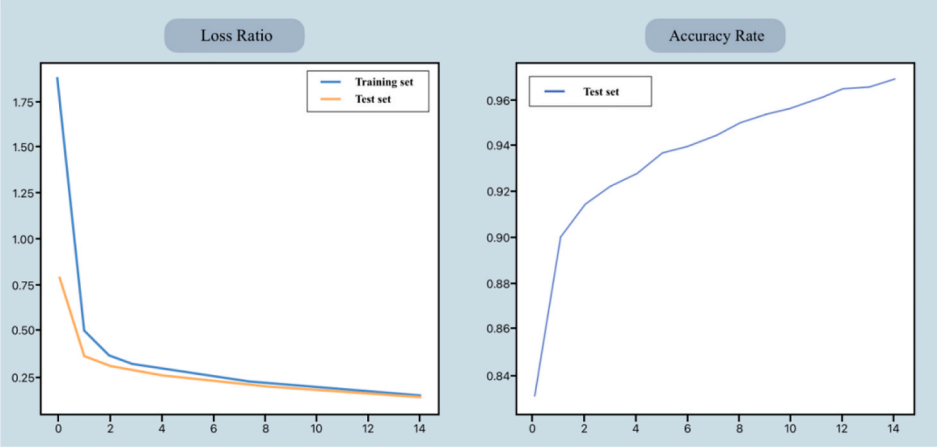


Fig. 20. Loss Function Curve of Network Model.

In this study, the model adopts two convolution layers, two pooling layers, and one fully connected layer to realize the fusion of infrared image and point cloud data. Fig. 5 illustrates the construction of the module's registration process using the PCRNet model. The model registered the transformed point cloud data with the field data collected through several iterations.

In the proposed PCRNet fusion model, infrared and point cloud data are defined as (64,64,3) RGB images as input data. There are 64 filters in the Infrared Data Convolution layer (Conv2D), whose convolution kernel size is defined as 3 × 3, and the ReLU activation function is used. The output of the first convolution layer is processed by the second convolution layer. The point cloud data processing layer is the same as the infrared data processing layer, and the point cloud data is processed through two convolution layers to extract the features of the point cloud data. Table 1 is PCRNet data fusion model brief code.

Two pooling layers are used to downsample the input infrared point cloud feature map to reduce the resolution of each feature map and effectively reduce the computation and storage requirements. Finally, the features of infrared and point cloud data are combined to form a larger feature vector. Through a fully connected layer, the number of neurons in this study was set to 128 and the ReLU function was used. In the data output phase, the sigmoid function is used to perform binary classification tasks. Finally, the PCRNet model performs data fusion by combining infrared and point cloud data features.Fig. 6 shows the 3D thermal image output after fusion of infrared thermal image and point cloud data.

3.2. Anomaly DetectionAlgorithm based on CNN

The anomaly automatic detection model based on CNN proposed in this study is used for anomaly detection of regular and irregular geometric envelope structures. By extracting temperature information and comparing it with the labels in the real abnormal data set, the detection accuracy, recall rate and precision of the evaluation model can be improved. The network architecture consists of five parts: input layer, convolutional layer, pooling layer, activation function layer, and full connection layer. Fig. 7 illustrates the specific network architecture. When analyzing image and video data, the simple pre-processed image and video data can be directly input into the convolutional neural network, and then the required features are extracted through the convolutional pooling process. The feature extraction results pass through the fully connected layer to obtain the final classification results. The essence of this process lies in the automatic learning of hierarchical features within images through convolution operations, progressing from low-level edge and texture characteristics to high-level semantic attributes (such as "hot spots" and "regions with abnormal temperature gradients"). Ultimately, it aims to classify each region as either "normal" or "abnormal.".

The first layer of convolution in the model will learn simple temperature gradient edges, while deeper convolutions will combine these basic features into more complex patterns, such as "a small high-temperature rectangular area" (corresponding to the outdoor unit of an air conditioner) or "a linear high-temperature band" (corresponding to a structural thermal bridge). The input data is compressed into low dimensional space by convolutional layer and pooling layer, and the key features of the data are extracted. Input data is reconstructed through up-sampling and convolution layers. The goal is to reconstruct normal data as closely as possible to the input. For normal data, the model can reconstruct the input image well. For abnormal data, the reconstruction error is large. Therefore, the image recognition result is classified by Softmax function to determine whether it is abnormal. Eq. (5), (6) is the Softmax loss function formula.

$$P_j^i = \frac{e^{a_j^{(i)}}}{\sum_{l=0}^{M-1} e^{a_j^{(i)}}} \# \tag{5}$$

$$Loss_{softmax} = -\frac{1}{N}\left[\sum_{i=1}^{N}\sum_{j=0}^{M-1} l\{y^{(i)} = j\} log P_j^i\right] \# \tag{6}$$

Where, the training sample is $\{x_i(x_i, y_i); i \in 1, \cdots, N; y \in 0, \cdots, M-1\}$; x_i represents the first input sample and y_i is the category label for that sample.

3.3. Reverse digital twin mapping

This study adopted a deep learning-based defect detection method to identify defects and extract their features from UAV images. The region convolutional neural network (R-CNN) developed based on an object instance segmentation framework is called Mask R-CNN. The abnormal defect marker pickup is mapped into the Digital Twin model using Mask R-CNN, including defect type, location, and quantity. Fig. 8 illustrates the workflow of Mask R-CNN. The workflow of mask R-CNN is similar to Faster R-CNN. It first extracts image features through multiple CNN layers, then trains a region proposal network (RPN) to generate a region proposal, and then generates a region of interest (RoI). The primary distinction lies in the addition of a branch to the Mask R-CNN, which predicts the split mask for each RoI in a manner similar to the current classification and bounding box regression processes. The mask R-CNN identifies each defect detection result with a bounding box that indicates its type and general location, reflecting its characteristics. Finally, the Digital Twin model integrates the identified defect data after coordinate transformation.

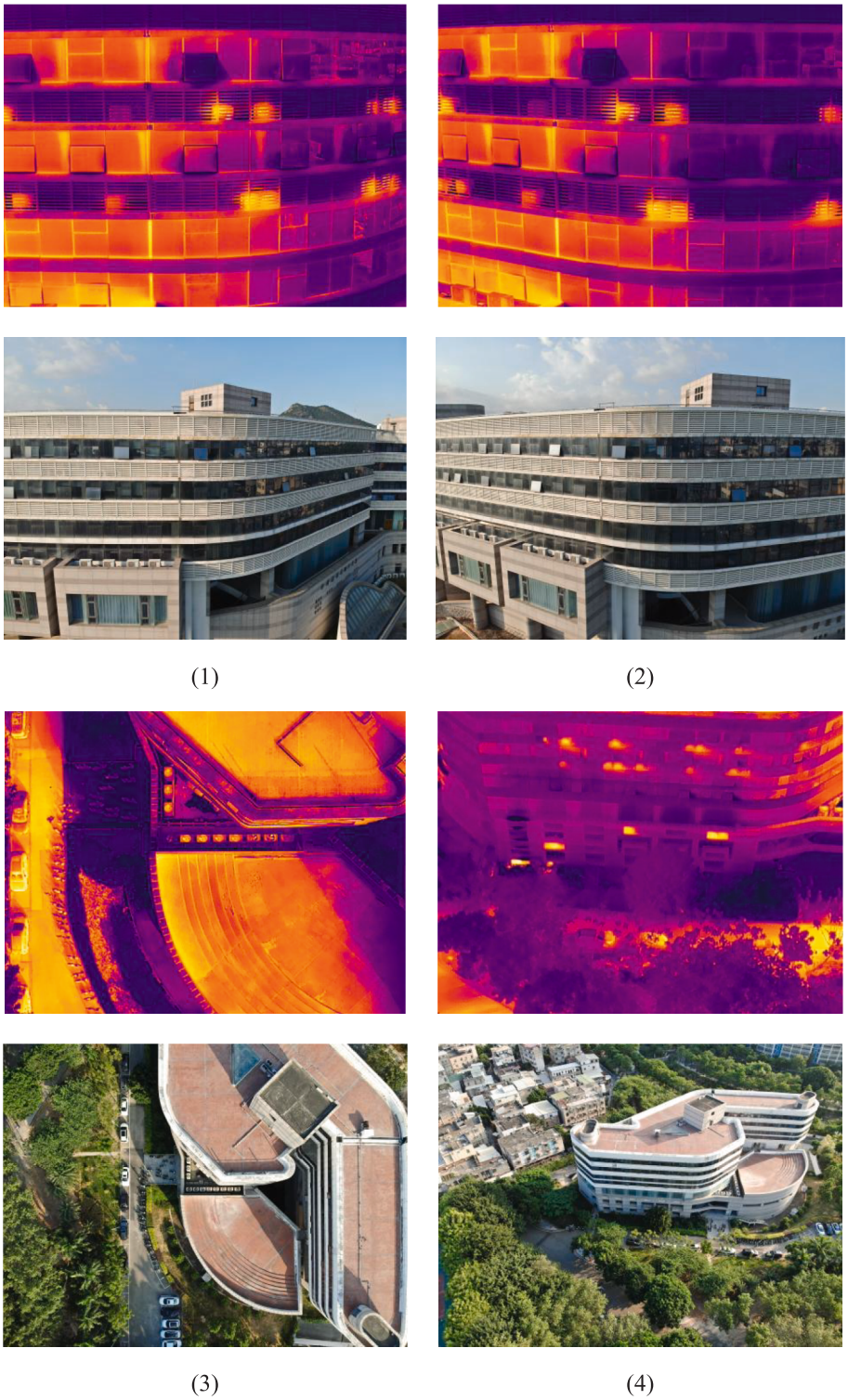


Fig. 21. Infrared Point Cloud Fusion Data (top) and Corresponding Positions (bottom).

In general, the specific method of the digital twin reverse mapping framework is to carry out abnormal detection of the building envelope through an anomaly detection algorithm, identify and mark abnormal parts and temperature information of the building envelope, and load the information into the digital twin model through digital twin reverse mapping technology. Fig. 9 illustrates the general framework for reverse digital twin mapping and managing anomaly information of the spatial geometric envelope, which relies on an infrared point cloud.

To sum up, the main workflow of the anomaly automatic detection model based on PCRNet-CNN proposed in this study is as follows:

(1) First, a higher-resolution infrared image is obtained by fusing it with the RGB image. Then, the infrared thermal image and its temperature and width bars are subjected to the same grayscale processing to obtain the grayscale image of the infrared thermal image and its temperature and width bars. Then, based on the

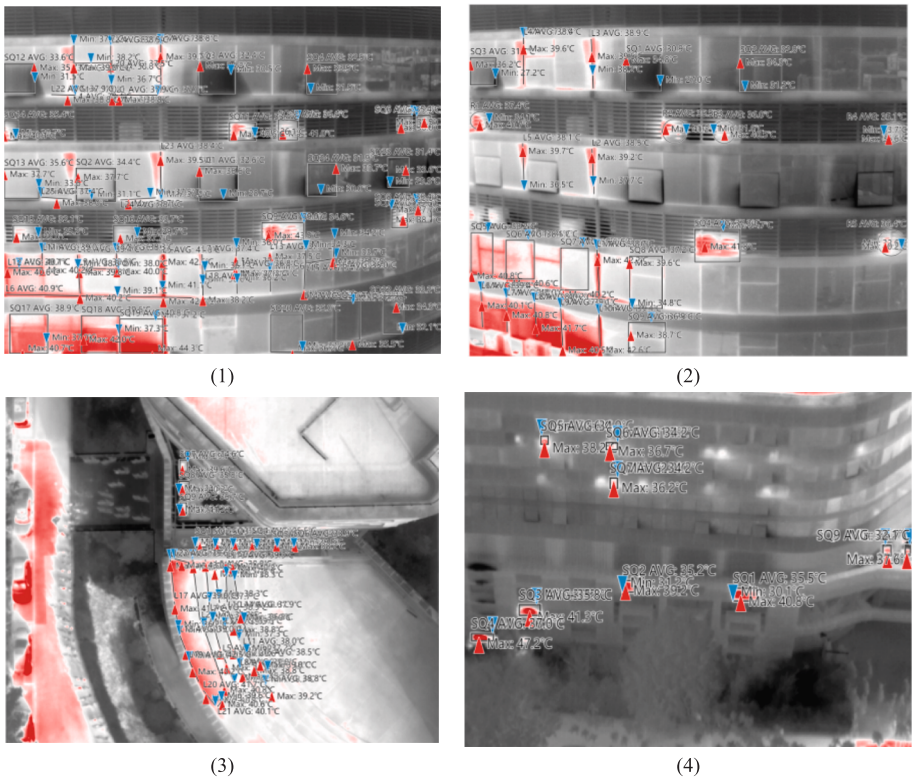


Fig. 22. Temperature Anomaly Detection Mark Diagram.

infrared thermal image and its grayscale image of the temperature width bar, interpolation processing is carried out on each pixel in the infrared thermal image to extract the temperature values of each pixel in the infrared thermal image. Secondly, the temperature values of each pixel in the extracted infrared thermal image are calibrated and compensated to obtain the temperature values of each pixel in the infrared thermal image with higher accuracy. Mark and classify abnormal points or areas. The algorithm is trained through the data in the database (the heat map data is general plane structures such as rectangles, rhombuses, etc., rather than arc-shaped planes, spherical planes, etc.). Finally, the developed algorithm is trained by manually identifying and marking the infrared images of the unmanned aerial vehicle mobile phone as the training database.

(2) Fuse the infrared thermal map of the non-conventional planar structure with the point cloud data to add depth information to it. Thus, detection can be carried out in the form of 3D reconstruction by using the CNN-based algorithm. Fuse two-dimensional images with three-dimensional point clouds to form a three-dimensional spatial model with depth information. The generated point cloud model can contain geometric, color, intensity and temperature information. Automate the marking of abnormal temperature locations in buildings to achieve the detection of abnormal building information.

(3) 3) Based on the point cloud model, establish the current situation model, and then integrate the point cloud and infrared information with the 3D model to create a systematic and intuitive 3D infrared point cloud model.

The next section will conduct an instance verification of the proposed automatic anomaly detection model to verify its accuracy and efficiency.

4. Case study

4.1. UAV data acquisition

In this study, Yinwu Building of Huaqiao University in Xiamen City, Fujian Province was selected as an experimental case to verify the anomaly detection algorithm of this study. The building envelope structure in this case includes: spherical, curved and other irregular structures, and vertical rectangular regular structures, as shown in Fig. 10. To verify whether the PCRNet infrared point cloud data fusion model and CNN anomaly detection model proposed in this study have good detection efficiency and robustness.

In this case, DJI M600 UAV was used for point cloud data acquisition, and the acquisition time was 14:00–18:00 with sufficient sunshine. The DJI Pilot set the flight path of the drone and flew five routes from different angles. As shown in Fig. 11, the drone fires vertically down and tilts at an Angle (left or right) toward the center of the measured area. The flight time of each route is about 7 min and 30 s. After flying 5 routes, all the construction site information is collected and the real three-dimensional point cloud model is generated, which takes about 37 min.

At the same time, the M600 is equipped with an infrared thermal imager ZENMUSE XT2 to fully scan the building to obtain its infrared thermal imaging data, and ensure that the data covers all parts of the building, including the exterior walls, roofs, Windows, etc., and is collected at different times in different weather conditions to obtain data under different light and temperature. Fig. 12 shows DJI UAV M600 and ZENMUSE XT2 respectively, Table 2 shows DJI UAV M600 parameters, Table 3 shows ZENMUSE XT2 parameters.

Fig. 13 shows the point cloud data collected by M600 UAV in this experiment and the infrared thermal imaging images under different weather conditions and different lighting conditions.

Table 6
Statistics of Abnormal Temperature Detection Information. (1).

Monitoring point	Minimum temperature (℃)	Average temperature (℃)	Maximum temperature (℃)
SQ1	28.7	32.6	36.6
SQ2	31.1	34.4	37.7
SQ3	30.5	32.4	34.2
SQ4	31.5	32.5	33.5
SQ5	29.8	32.9	36.0
SQ6	33.2	35.4	37.5
SQ7	32.5	35.6	38.7
SQ8	33.0	35.4	37.7
SQ9	34.6	39.0	43.5
SQ10	31.8	36.4	41.0
SQ11	26.1	33.4	40.6
SQ12	31.5	33.6	35.7
SQ13	33.6	35.6	37.7
SQ14	30.7	35.4	40.1
SQ15	25.2	32.1	39.0
SQ16	29.7	33.7	37.7
SQ17	37.0	38.9	40.7
SQ18	37.3	39.6	42.0
SQ19	37.2	40.8	44.3
SQ20	33.8	35.0	36.2
L1	37.8	38.8	39.7
L2	38.2	38.5	38.8
L3	38.2	38.7	39.2
L4	37.7	38.4	39.0
L5	41.1	41.8	42.5
L6	39.1	40.9	42.7
L7	36.6	37.4	38.2
L8	36.5	38.3	40.0
L9	38.0	38.9	39.8
*L*10	38.6	39.4	40.2
*L*11	37.7	39.0	40.2
L12	39.7	40.1	40.6
L13	36.7	37.1	37.5
L14	34.3	35.0	35.8
L15	33.7	34.8	35.8
L16	34.2	35.0	35.7
L17	36.0	37.4	38.8
L18	36.7	37.3	37.8
L19	36.2	36.8	37.3
L20	36.7	37.5	38.2
L21	37.0	37.9	38.8
L22	37.0	37.9	38.8
L23	37.2	38.4	39.5
L24	37.5	38.1	38.7
L25	36.7	37.4	38.0

Table 7
Statistics of Abnormal Temperature Detection Information. (2).

Monitoring point	Minimum temperature (℃)	Average temperature (℃)	Maximum temperature (℃)
SQ1	27.0	30.9	34.8
SQ2	31.2	32.8	34.3
SQ3	27.2	31.7	36.2
SQ4	32.7	37.3	41.8
SQ5	35.6	38.2	40.8
SQ6	35.7	38.1	40.6
SQ7	35.0	37.6	40.2
SQ8	34.8	37.2	39.6
SQ9	34.3	36.5	38.7
R1	34.1	37.4	40.7
R2	31.7	36.5	41.2
R3	31.6	36.0	40.5
R4	33.7	35.1	36.6
R5	32.6	36.4	40.2
L1	35.2	38.7	42.2
L2	37.7	38.5	39.2
L3	38.2	38.9	39.6
L4	37.2	38.4	39.6
L5	36.5	38.1	39.7
L6	38.6	39.4	40.1
L7	39.2	40.0	40.8
L8	38.7	40.2	41.7
L9	37.7	39.1	40.5
*L*10	36.2	39.4	42.6

4.2. UAV data processing and fusion

4.2.1. Transformation and processing of point cloud data

The processed oblique photography point cloud data is imported into DJI Terra software. To perform the import operation, navigate to the "File" menu bar, select "Import Data", and then select the folder containing the imported tilt photography point cloud data. After the data is imported, it is preprocessed by DJI Terra, including data correction, registration, and ghosting removal. DJI Terra then carries out feature extraction and matching, extracting feature points from oblique photographic images and using these points for image registration and matching. Following feature matching and registration, we convert the matched pixel points into three-dimensional points based on the geometric information of the image and camera parameters. The converted point cloud data is shown in Fig. 14.

After building the point cloud model, CloudCompare software removes point clouds irrelevant to the case and reduces the number of points in the point cloud model through uniform filtering. The point cloud simplification is quickly completed by CloudCompare, as shown in Fig. 15.

4.2.2. Infrared image Mass processing

Noise reduction of infrared data is carried out by Gaussian filtering method. Gaussian filter can effectively eliminate Gaussian noise according to the features of the image, and retain the edge features of the image. The infrared image data after processing in this way can more clearly show the details and features of the image, and provide a better basis for the subsequent data processing. Fig. 16 shows the infrared image after Gaussian filtering.

4.2.3. Data fusion

The PCRNet data fusion model proposed in this study fuses infrared image with point cloud model. First, the features in the original infrared image are extracted, and the features in the point cloud data are extracted and fused by combining the features of the infrared and point cloud. The fused data is shown in Fig. 17.

4.3. CNN model detection recognition and labeling

The real building maintenance anomaly data set will be collected, including infrared point cloud data and corresponding anomaly markers.Depending on the actual situation, the exceptions may include energy waste, equipment failure, and security risks. Ensure that the data set covers all types of exceptions. DJI Thermal Analysis Tool 3 generated the data set. DAT3 imported the infrared thermal imaging data of the building envelope. Use the annotation tool in the DAT3 toolbar to label abnormal areas in the infrared thermal imaging images. Anomalous regions may appear as abnormal temperatures or uneven energy distribution. The abnormal area is labeled by using the rectangle annotation tool or the free drawing tool, as shown in Fig. 18, and its temperature information (SQ1-24, L1-25) is shown in Table 4.

By comparing the minimum, maximum and average temperature of the abnormal area with the temperature information of the surrounding normal area, the reason of the abnormal building envelope structure is determined by field investigation. The temperature information can also be used to determine whether there is a labeling error. For example, the highest temperature of SQ19 is 44.3℃, and the lowest temperature is 37.2℃. Combined with the infrared point cloud thermal image, it can be determined that this position is the temperature emitted by the external unit of the air conditioner. By combining infrared temperature

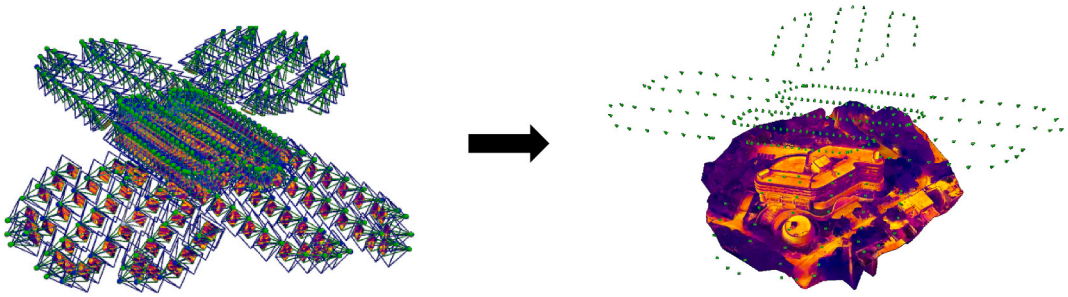


Fig. 23. Digital Twin Reverse Mapping.

Table 8
Temperature Anomaly Detection Statistics. (3).

Monitoring point	Minimum temperature (℃)	Average temperature (℃)	Maximum temperature (℃)
SQ1	29.8	36.3	42.7
SQ2	28.2	35.1	42.0
SQ3	27.2	34.0	40.8
SQ4	29.7	35.5	41.2
SQ5	29.7	34.2	38.7
SQ6	29.5	33.9	38.2
SQ7	29.7	34.6	39.6
SQ8	31.3	35.8	40.2
SQ9	32.2	36.7	41.2
L1	37.6	40.0	42.5
L2	37.6	39.2	40.8
L3	37.7	39.5	41.2
L4	37.6	38.9	40.2
L5	39.5	40.0	40.5
L6	38.3	38.8	39.3
L7	39.0	39.3	39.6
L8	39.3	39.8	40.2
L9	37.5	38.8	40.0
L10	37.3	38.0	38.8
L11	37.2	38.0	38.8
L12	38.3	38.8	39.2
L13	38.3	39.3	40.2
L14	36.8	37.9	39.0
L15	38.1	38.5	39.0
L16	38.2	41.6	45.1
L17	36.2	39.0	41.7
L18	36.0	39.0	42.1
L19	41.7	42.5	43.2
L20	40.3	41.7	43.1
L21	39.6	40.1	40.6
L22	36.3	39.6	43.0

information in this way, the detection speed of building envelope and the detection accuracy of evaluation model can be greatly improved.

4.3.1. Model training and evaluation index

The hierarchical random sampling strategy is adopted to divide the data set of this model. The training set (70 %) covers typical areas such as building facades and roofing, and contains infrared point cloud fusion data of different geometric complexity (regular rectangle and curved surface) to ensure that the model learns diversified defect features. Test set (30 %): Independent of the training set, contains data on building areas not involved in training and is used to assess the generalization ability of the model. The parameters used in CNN model training are shown in Table 5.

Fig. 19 is the training flow chart of CNN abnormal defect detection model in this study.

The trained CNN model is applied to the collected building infrared point cloud data set for anomaly detection. Record the anomalies detected by the model and extract the temperature information. Compared with the marks in the real anomaly data set, the model's detection accuracy rate, recall rate, and accuracy rate are evaluated. The

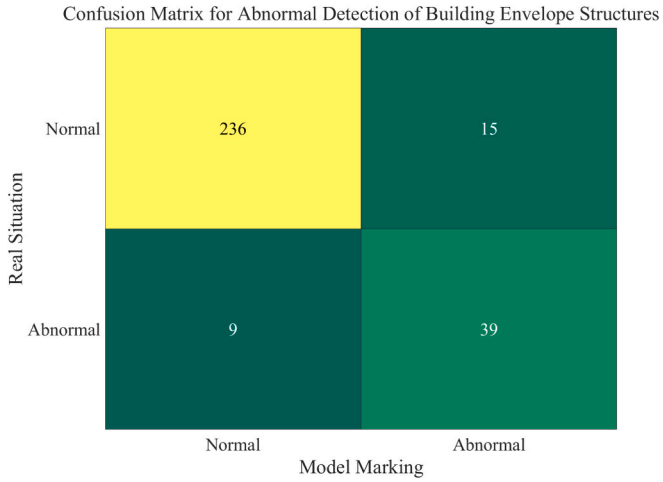


Fig. 24. Confusion Matrix for Abnormal Detection of Building Envelope Structures.

Table 9
Statistics of Abnormal Temperature Detection Information. (4).

Monitoring point	Minimum temperature (℃)	Average temperature (℃)	Maximum temperature (℃)
SQ1	30.1	35.5	40.8
SQ2	31.2	35.2	39.2
SQ3	30.2	35.8	41.3
SQ4	26.8	37.0	47.2
SQ5	31.6	34.9	38.2
SQ6	31.7	34.2	36.7
SQ7	32.2	34.2	36.2
SQ8	30.7	34.2	37.7
SQ9	26.7	32.1	37.6

Table 10
Hidden Danger Detection Time of Different Items.

Detection area	Hidden danger detection label type	Accuracy rate	Hazard detection time (s)
(1)	Rectangular mark	0.91	6.06
	Circular mark	0.89	
	Linear label	0.95	
(2)	Rectangular mark	0.93	5.07
	Circular mark	0.88	
	Linear label	0.86	
(3)	Rectangular mark	0.92	5.21
	Circular mark	–	
	Linear label	0.94	
(4)	Rectangular mark	0.96	4.51
	Circular mark	–	
	Linear label	–	
Mean value		0.92	5.21

Table 11
Case Evaluation Results.

Precision	Recall Rate	F1-Score
96.3 %	94 %	95.1 %

anomaly detected by the model is compared with the real mark to determine the correctness and accuracy of the detection.

The evaluation indicators include Overall Accuracy (OA), Mean Accuracy (mAcc) and mean Intersection over-Union (mIoU), which are represented by Eq. (7), (8), (9) respectively.

$$OA = \frac{\sum_{i=1}^{c} TP}{\sum_{i=1}^{c} (TP + FN)} \# \tag{7}$$

$$mAcc = \frac{1}{c} \bullet \sum_{i=1}^{c} \frac{TP}{TP + FN} \# \tag{8}$$

$$mIoU = \frac{1}{c} \sum_{I=1}^{C} \frac{TP}{TP + FN + FP} \# \tag{9}$$

Where, TP (True Positive) indicates that the prediction result of the classifier is positive sample; FP (False Positive) indicates that the classifier predicts a positive sample; TN (True Negative) indicates that the classifier predicts a negative sample. FN (False Negative) means that the classifier predicts a negative sample; c represents the total number of categories.

The anomaly detection algorithm evaluates the network model through 14 iterations of training, and the accuracy rate of the network model obtained by the experiment is 95.8 %. The loss function curve is shown in Fig. 20.

4.4. Abnormal information detection

In order to verify the reliability of the anomaly detection algorithm, the accuracy and detection time were evaluated in this study. By comparing the temperature information of the building envelope in the experimental case and comparing it with the marks in the real anomaly data set, the location with large temperature difference in the envelope is identified, and the abnormal location is marked by the CNN anomaly information detection model, so as to reduce the abnormal detection of the whole building by inspectors. As shown in Fig. 21, this is the data that was used to make sure that the anomaly detection algorithm worked correctly. It is a combination of data from the envelope structure of different parts of the Yinwu building. The data above is infrared point cloud fusion data, and the data below shows where it was in the real scene.

The CNN anomaly detection model proposed in this study is used to detect anomalies in the selected data, as shown in Fig. 22, and its information statistics are shown in Tables 6, 7, 8, 9.

4.5. Digital twin reverse mapping

The detection information is registered to the digital twin model through the coordinate mapping relationship, as shown in Fig. 23. This integration of the location and type of defect information with the digital twin model facilitates the visualization and management of defects for subsequent maintenance. Data management in a digital twin environment is systematic and intuitive. Provides visual interaction for data users.

In Fig. 22 it can be seen that the infrared point cloud data of the abnormal area of the envelope structure detected by this research were reverse-mapped by digital twins, and Mask R-CNN was used to pick up the abnormal defect location markers and temperature information into the digital twin model. Through the general framework of defect detection and management of building envelope proposed by the research, users can carry out systematic and intuitive data management in the digital twin environment, and provide visual interaction.

4.6. Result and Discussion

Through the abnormal detection mark and combined with the actual situation, it is found that the abnormal parts of the case project mostly occur in the curtain wall of the project. For example, as shown in Fig. 22 ((1) and (2)) of the abnormal detection results, the red rectangle or circle is marked as the temperature in the high temperature area is about 40 °C (with a larger deviation from the surrounding temperature). Combining the actual information reveals that the abnormal part is not the curtain wall's defective part, but rather the air conditioner's external unit, which the high temperature has identified and marked. However, the abnormal areas marked by the straight line in the abnormal detection mark mostly occur at the connection of the curtain wall. According to the field investigation of the building, the following problems are found, such as damage to thermal insulation materials, poor sealing, and structural defects. As for the inspection Fig. 22 ((3)) of the building roof, the abnormal parts mostly occur in the joint seal parts. Such structural parts will produce a thermal bridge effect, resulting in uneven heat transfer and resulting in structural damage, which requires special attention. In addition, the indoor beam, plate, and column junction is also a place where thermal bridge effect occurs frequently, which requires special attention.

The anomaly detection algorithm developed in this research can quickly complete the anomaly detection for a single set of data. Table 8 displays the detection time for other hidden dangers, demonstrating an average accuracy of 0.92 and a detection speed of 5.21 s. According to the results, the project tested a total of 299 items of data, and it took about 26 min to complete the abnormal detection of the building. In addition, the anomaly detection in this study was carried out through the comparison of the surface temperature of the outer wall envelope structure, and each data collection time was short, so it would not be affected by the weather and orientation, nor by the data of different time periods.

Compared with the conclusions of existing abnormal defect detection studies, Wang et al. [50] adopted the defect detection model based on YOLOv5 to achieve an accuracy of 82.41 %. Li et al. [51] integrated 2D plane features and 3D voxel features in GPRB scanning to achieve a detection rate of 62 % and 33 % in voids and cracks respectively. Mariluz et al. [52] proposed to combine UAV infrared thermal imaging and photogrammetry technology to detect thermal defects in building envelope structures. This study can have high accuracy in common envelope structures, but high identification error in complex geometric structures. Through comparison, the accuracy of CNN anomaly detection model proposed in this study is better than that of previous studies. And the response speed has also improved.

To comprehensively evaluate the performance and effect of the scheme, this paper collates and analyzes 299 available data items and draws a confusion matrix for abnormal detection of building envelope structures, as shown in Fig. 24.

Based on the data in the figure, the Precision, Recall Rate and F1-Score of the research case results can be calculated, as shown in Table 9.Table 10.Table 11..

It can be analyzed from Table 9 that the proposed scheme has high performance and practicality, with excellent accuracy and recall rate. Even with high Precision detection results, it maintains a high Recall Rate, ensuring that serious safety hazards (such as structural cracking and severe failure of the insulation layer) are almost impossible to be missed, greatly reducing the risk of safety accidents caused by missed detection. Furthermore, the 95.1 % F1 score reflects that the comprehensive performance of the model is very balanced and stable, reaching the standard for practical engineering applications, and also proves the feasibility and efficiency of the proposed algorithm framework model.

5. Conclusion and future studies

In this study, the application of infrared point cloud fusion technology, deep learning, and BIM management in the defect detection of spatial geometric envelope structure is discussed in depth. In this paper, we propose a data fusion model based on PCRNet and introduce the PCA module of the Adam activation function, which makes the model training speed up and has stronger robustness.

An algorithm model based on CNN was established to detect the anomalies of regular and irregular geometric enclosures. UAV collected the infrared heat map and point cloud data of the building envelope, then processed the data through noise reduction, segmentation, and three-dimensional reconstruction. This allowed for the extraction of temperature information and its comparison with the marks in the real abnormal data set, thereby reducing human resource consumption and enhancing detection efficiency. After experimental verification, the developed anomaly detection algorithm can quickly complete the anomaly detection for a single set of data; the average accuracy rate of other hidden danger detection is 0.92, and the average detection speed is 5.21 s. Additionally, this study conducted anomaly detection by comparing the surface temperature of the outer wall envelope structure, ensuring that the data collection time remained consistent and unaffected by weather, orientation, or time constraints.

This research proposed a general framework for defect detection and management of building envelope structures based on the combination of infrared thermal imaging and point cloud, which can carry out systematic and intuitive data management in a digital twin environment. Enable users to visualize, query, and analyze temperature with reference to 3D spatial information, thereby speeding up the building diagnosis and modeling process. Provides visual interaction for data users.

However, the dataset of this study is relatively simple, with only the content of the dataset designed for a single case, which has certain limitations for research in other industries. Future research can collect building data from different climate zones, various building types (such as stadiums and airport terminals), different materials, and different degrees of degradation to construct a large-scale and diverse benchmark dataset, in order to train more universal and powerful models. In addition, current research mainly focuses on the detection of thermal defects (such as thermal Bridges and damage to insulation layers), and has limited recognition capabilities for other types of defects (such as fine cracks, surface weathering, and mold growth). Future research can conduct multi-modal fusion of various sensing data such as visible light, infrared, and hyperspectral data. Realize the leap from “ingle thermal anomaly detection” to “comprehensive building pathological diagnosis”.

Finally, the current data processing, analysis and mapping processes still require certain manual intervention and have not yet formed a fully end-to-end automated closed loop. We hope to deploy the algorithm on the cloud platform in future research. The data collected by the drones can be transmitted to the cloud in real time for processing and analysis, and the results can be fed back to users in near real time through the 5G network. This can greatly shorten the cycle from detection to decision-making. In the future, it is possible to explore the introduction of large language models to develop operation and maintenance decision-making agents. These agents can not only detect and label defects but also automatically analyze the causes of defects, assess risk levels, and even automatically estimate maintenance costs and construction periods, providing managers with deeper decision support.

CRediT authorship contribution statement

Wenjie Shen: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Chen Wang:** Writing – review & editing, Supervision, Software, Methodology, Data curation. **Can Zhu:** Visualization, Validation, Supervision, Methodology. **Defa Wang:** Validation, Supervision, Project administration, Data curation, Conceptualization. **Zhibin Chen:** Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors gratefully acknowledge the financial support provided by the Fujian Provincial Department of Science and Technology (Grant Number: 2021I0014), the Xiamen Municipal Construction Bureau (Grant Number: XJK2022-1-7), and the Fujian Provincial Department of Housing and Urban Rural Development (Grant Number: 2023-K-31; 2023-K-63; 2023-K-73).

Data availability

No data was used for the research described in the article.

References

[1] N.H. Pan, C.H. Tsai, K.Y. Chen, J. Sung, Enhancement of external wall decoration material for the building in safety inspection method, J. Civ. Eng. Manag. 26 (2020) 216–226.
[2] D.G. Lowe, Object recognition from local scale-invariant features, Proc of IEEE International Conference on Computer Vision (1999).
[3] S. Jung, S. Song, P. Youn, H. Myung, Multi-Layer Coverage Path Planner for Autonomous Structural Inspection of High-Rise Structures, IEEE, DOI (2018).
[4] S. Siebert, J. Teizer, Mobile 3D Mapping for surveying Earthwork using an Unmanned Aerial Vehicle (UAV). 30th International Symposium on Automation and Robotics in Construction and Mining; Held in Conjunction with the 23rd World Mining Congress, 2013.
[5] Z. Yang, W. Sun, F. Liu, Y. Zhang, X. Chen, Z. Wei, X. Li, Field collaborative recognition method and experiment for thermal infrared imaging of damaged potatoes, Comput. Electron. Agric. 223 (2024).
[6] P. Das, S. Nath, R. Gupta, S.D. Roy, M.K. Bhowmik, Infrared thermogram image guided discontinuous appearances of hyaluronic acid for classification of arthritic knee joints, J. Therm. Biol 123 (2024) 103915.
[7] H. Li, Y. Liu, H. Wu, B. Lin, L. Lei, J. He, Thermal preference prediction through infrared thermography technology: Recognizing adaptive behaviors, Build. Environ. 262 (2024) 111829.
[8] A. Kylili, P.A. Fokaides, P. Christou, S.A. Kalogirou, Infrared thermography (IRT) applications for building diagnostics: a review, Appl. Energy 134 (2014) 531–549.
[9] R. Vadivambal, D.S. Jayas, Applications of thermal Imaging in Agriculture and Food Industry—A Review, Food Bioproc. Tech. 4 (2011) 186–199.
[10] F. Li, D.J. Colvin, V.S. Pavan Buddha, K.O. Davis, G. Tamizhmani, Electroluminescence and infrared imaging of fielded photovoltaic modules: a complementary analysis of series resistance-related defects, Sol. Energy 276 (2024) 112704.
[11] Y.J. Cha, D.L. Gopal, R. Ali, Vision-based concrete crack detection technique using cascade features, Sensors & Smart Structures Technologies for Civil, Mechanical, & Aerospace Systems (2018).
[12] T. Omar, M. Nehdi, Condition Assessment of Reinforced Concrete Bridges: Current Practice and Research challenges, Infrastructures 3 (2018).
[13] S. Yehia, O. Abudayyeh, S. Nabulsi, I. Abdelqader, Detection of Common Defects in Concrete Bridge Decks using Nondestructive Evaluation Techniques, J. Bridg. Eng. 12 (2007) 215–225.
[14] S. Abu Dabous, S. Yaghi, S. Alkass, O. Moselhi, Concrete bridge deck condition assessment using IR Thermography and Ground Penetrating Radar technologies, Autom. Constr. 81 (2017) 340–354.
[15] X. Chen, G. Tian, J. Wu, C. Tang, K. Li, Feature-based Registration for 3D Eddy Current Pulsed Thermography, IEEE Sensors Journal, DO I (2019) 6998–7004.
[16] T. Wang, V.J.L. Gan, Enhancing 3D reconstruction of textureless indoor scenes with IndoReal multi-view stereo (MVS), Autom. Constr. 166 (2024) 105600.
[17] Y. Su, J. Wang, X. Wang, L. Hu, Y. Yao, W. Shou, D. Li, Zero-reference deep learning for low-light image enhancement of underground utilities 3D reconstruction, Autom. Constr. 152 (2023) 104930.
[18] Y. Huang, C.H. Chiang, K.T. Hsu, Combining the 3D model Generated from Point Clouds and Thermography to Identify the Defects Presented on the Facades of a Building, SPIE Smart Structures and Materials + Nondestructive Evaluation and Health Monitoring Conference (2018).
[19] L. Hoegner, U. Stilla, Mobile thermal mapping for matching of infrared images with 3D building models and 3D point clouds, Quantitative Infrared Thermography Journal, DO I (2018) 1–19.
[20] C.A. Balaras, A.A. Argiriou, Infrared thermography for building diagnostics, Energy & Buildings 34 (2002) 171–183.

[21] J. Feng, J. Li, K. Gao, G. Wu, J. Tian, Y. Zou, Portable automatic detection system with infrared imaging for measuring steel wires corrosion damage, Autom. Constr. 156 (2023).
[22] R.W. Martin, A. Sabato, A. Schoenberg, R.H. Giles, C. Niezrecki, Comparison of nondestructive testing techniques for the inspection of wind turbine blades' spar caps, Wind Energy 21 (2018) 980–996.
[23] F. Liu, J. Liu, L. Wang, Asphalt pavement fatigue crack severity classification by infrared thermography and deep learning, Autom. Constr. 143 (2022) 104575.
[24] E. Ichi, S. Dorafshan, Effectiveness of infrared thermography for delamination detection in reinforced concrete bridge decks, Autom. Constr. 142 (2022) 104523.
[25] N.P. Avdelidis, A look on thermography: from passive to active NDT &, E Surveys (2007) 654115.
[26] W. Guoqiang, Z. Hongxia, G. Zhiwei, S. Wei, J. Dagong, Bilateral filter denoising of Lidar point cloud data in automatic driving scene, Infrared Phys. Technol. 131 (2023).
[27] H. Si, X. Wei, Feature extraction and representation learning of 3D point cloud data, Image Vis. Comput. 142 (2024).
[28] Y.K. Cho, M. Gai, Projection-Recognition-Projection Method for Automatic Object Recognition and Registration for Dynamic Heavy Equipment Operations, J. Comput. Civ. Eng. 28 (2014) A4014002.
[29] T. Kaiser, C. Clemen, H.G. Maas, Automatic co-registration of photogrammetric point clouds with digital building models, Automation in construction, DOI, 2022.
[30] L. Truong-Hong, R. Lindenbergh, Extracting structural components of concrete buildings from laser scanning point clouds from construction sites, Adv. Eng. Inf. 51 (2022) 101490-.
[31] M.K. Kim, J.P. Putrathedja, Q. Wang, Automated dimensional quality assessment for formwork and rebar of reinforced concrete components using 3D point cloud data, Automation in Construction, 112 (2020) 103077-.
[32] J. Dong, N. Wang, H. Fang, H. Lu, D. Ma, H. Hu, Automatic augmentation and segmentation system for three-dimensional point cloud of pavement potholes by fusion convolution and transformer, Adv. Eng. Inf. 60 (2024).
[33] M. Puliti, G. Montaggioli, A. Sabato, Automated subsurface defects' detection using point cloud reconstruction from infrared images, Autom. Constr. 129 (2021) 103829.
[34] P. Stałowska, C. Suchocki, M. Rutkowska, Crack detection in building walls based on geometric and radiometric point cloud information, Autom. Constr. 134 (2022) 104065.
[35] S.A. Rahaman, T. Urmee, D.A. Parlevliet, Investigate the impact of environmental and operating conditions of infrared (IR) imaging on infrared thermography of PV modules to identify defects, Sol. Energy 245 (2022) 231–253.
[36] Z. Ma, S. Liu, A review of 3D reconstruction techniques in civil engineering and their applications, Adv. Eng. Inf. 37 (2018) 163–174.
[37] R. Ali, J. Zeng, M. Kavgic, Y.J. Cha, Heat loss detection using thermal imaging by a small UAV prototype, DOI (2020).
[38] Y. Li, J. Zou, J. Jing, H. Jin, H. Yu, Establish keypoint matches on multispectral images utilizing descriptor and global information over entire image, Infrared Physics & Technology, DO I (2016) 1–10.
[39] C. Aguilera, F. Barrera, F. Lumbreras, A.D. Sappa, R. Toledo, Multispectral image Feature Points, Sensors 12 (2012) 12661–12672.
[40] S. Dabetwar, N.N. Kulkarni, M. Angelosanti, C. Niezrecki, A. Sabato, Sensitivity analysis of unmanned aerial vehicle-borne 3D point cloud reconstruction from infrared images, Journal of Building Engineering 58 (2022) 105070.
[41] A. Di Tommaso, A. Betti, G. Fontanelli, B. Michelozzi, A Multi-Stage model based on YOLOv3 for defect detection in PV panels based on IR and Visible Imaging by Unmanned Aerial Vehicle, arXiv e-prints, DOI (2021).
[42] A. Prasanth, M. Sundarrajan, C.M. Deepak, J. Akshya, Enhancing security of medical images using code-based intermittent encryption and convolutional neural network, The Computer Journal 5 (2024) 5.
[43] Zaken, H. A. , Prasanth, A. , & Jahjah, M. A. . An Automated Deep Learning-Based VAR for Enhancing Referee Decision-Making in Football. #i{2025 3rd International Conference on Intelligent Data Communication Technologies and Internet of Things (IDCIoT)}. IEEE.
[44] Amiri, Z. , Heidari, A. , Navimipour, N. J. , Esmaeilpour, M. , & Yazdani, Y. . (2024). The deep learning applications in iot-based bio- and medical informatics: a systematic literature review. #i{Neural Computing & Applications}#i{,} #i{36} (11).
[45] N. Sang, T. Zhang, Segmentation of FLIR images by Hopfield neural network with edge constraint - ScienceDirect, Pattern Recogn. 34 (2001) 811–821.
[46] Y.J. Cha, W. Choi, O. Büyüköztürk, Deep Learning-based Crack damage Detection using Convolutional Neural Networks, Computer-Aided Civil and Infrastructure Engineering, DOI, 2017.
[47] Q. Fang, B.D. Nguyen, C.I. Castanedo, Y. Duan, X. Maldague, Automatic defect detection in infrared thermography by deep learning algorithm, Conference on Thermosense: Thermal Infrared Applications (2020).
[48] Y. Duan, S. Liu, C. Hu, J. Hu, J. Meng, Automated defect classification in infrared thermography based on a neural network, NDT & E International, 107 (2019) 102147.
[49] R. Ali, Y.J. Cha, Subsurface damage detection of a steel bridge using deep learning and uncooled micro-bolometer, Constr. Build. Mater. 226 (2019) 376–387.
[50] Wang, J. , WeiWang, Zhang, Z. , Lin, X. , Zhao, J. , & Chen, M. , et al. (2024). Yolo-dd: improved yolov5 for defect detection. Computers, Materials & Continua, 78(1).
[51] H. Li, N. Li, R. Wu, H. Wang, D. Song, Gpr-rcnn: an algorithm of subsurface defect detection for airport runway based on gpr, IEEE Rob. Autom. Lett. PP(99) (2021) 1.
[52] M. Gil-Docampo, J.O. Sanz, I.C. Guerrero, Manés Fernández Cabanas, Uas ir-thermograms processing and photogrammetry of thermal images for the inspection of building envelopes, Appl. Sci. 2023 (2076-3417), 13(6).